The Triadic Cycle with AdS/CFT
Recursion, Emergent Time, and Holographic Structure

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June 2026

Abstract

This paper develops a dynamical and holographic refinement of the Triadic Cycle, a minimal architectural structure linking energy, geometry, and information through the closed sequence $E \to G \to I \to E$. Building on earlier work that established the Triadic Cycle as a theory-independent ontology underlying modern physics, the present paper introduces a formal recursive interpretation of the cycle and identifies recursion as the generator of emergent temporal structure. Time is treated not as a fundamental parameter but as an iterative ordering induced by successive triadic updates, reflecting the reorganization of informational, energetic, and geometric roles.

A central contribution of this work is the explicit embedding of the Triadic Cycle within the AdS/CFT correspondence. We show that the recursive index of the triadic update can be interpreted as a holographic renormalization direction, that informational and geometric roles map naturally onto entanglement entropy and bulk extremal surfaces, and that energetic roles correspond to boundary Hamiltonian evolution. This mapping clarifies how the Triadic Cycle is instantiated within a mainstream theoretical framework and why holographic dualities satisfy the architectural constraints derived in previous Triadic Cosmos papers.

The triadic framework further yields a structurally clean interpretation of black-hole interiors. Because the interior lacks independent informational, energetic, and geometric degrees of freedom, the triadic recursion terminates at the horizon. As a consequence, the interior does not support emergent temporal structure and is best understood as a holographic shadow of boundary data rather than a dynamical region. This resolves classical singularities, avoids information loss, and eliminates the need for firewalls or exotic microstate geometries.

The resulting synthesis provides a concrete dynamical interpretation of the Triadic Cycle, clarifies the structural origin of emergent time, and positions AdS/CFT as a canonical realization of triadic interdependence. The paper concludes by outlining structural predictions, conceptual implications, and open questions for future mathematical development of the Triadic Cosmos framework.

Contents

1 Introduction

Modern physics exhibits a remarkable degree of empirical success while remaining conceptually fragmented. Quantum mechanics, quantum field theory, general relativity, thermodynamics, and holographic dualities each illuminate essential aspects of physical law, yet they do so using distinct structural assumptions and incompatible ontological commitments. Attempts at unification typically introduce new mathematical frameworks or microscopic models, but these efforts often inherit the limitations of the theories they aim to reconcile. What is missing is a structural account of the roles that physical theories must instantiate in order to remain mutually compatible.

The Triadic Cosmos program has proposed such an account. Earlier work introduced a minimal ontology consisting of three irreducible roles—energy, geometry, and information—and showed that modern physics already implements a closed interdependence among them, expressed as the Triadic Cycle $E \to G \to I \to E$. Subsequent papers developed the architectural consequences of this ontology, identifying theory-independent constraints that any viable unifying framework must satisfy. These works established the Triadic Cycle as a structural grammar underlying quantum theory, relativity, thermodynamics, and holography, but they remained architectural rather than dynamical: the cycle was treated as a conceptual structure rather than as an explicit process.

The present paper advances the Triadic Cosmos framework by introducing a formal dynamical interpretation of the Triadic Cycle. We treat the cycle as a recursive update scheme in which each triadic role is iteratively reorganized by the others. This recursion provides a natural generator of emergent temporal structure: time is interpreted not as a fundamental parameter but as the ordering induced by successive triadic transformations. In this view, temporal evolution reflects the iterative reconfiguration of informational, energetic, and geometric roles, and equilibrium corresponds to fixed points of the recursive process.

A second and central contribution of this work is the explicit embedding of the Triadic Cycle within the AdS/CFT correspondence. Holographic dualities provide the most concrete realization of the deep relationship between geometry and information, and the recursive structure of holographic renormalization offers a natural setting in which to interpret triadic recursion. We show that the recursive index of the Triadic Cycle can be understood as a holographic renormalization direction, that informational roles map onto entanglement entropy and boundary degrees of freedom, that geometric roles correspond to bulk extremal surfaces, and that energetic roles align with boundary Hamiltonian evolution. This mapping clarifies why holographic frameworks satisfy the architectural constraints derived in earlier Triadic Cosmos papers and positions AdS/CFT as a canonical realization of triadic interdependence.

A third contribution of this paper is a structural interpretation of black-hole interiors that emerges naturally from the triadic and holographic framework. Because the interior of a black hole lacks independent informational, energetic, and geometric degrees of freedom, the triadic recursion terminates at the horizon. As a consequence, the interior does not support emergent temporal structure and is best understood as a holographic shadow of boundary data rather than a dynamical region. This perspective resolves classical singularities, avoids information loss, and eliminates the need for firewalls or exotic microstate geometries, providing a clean and architecturally coherent account of black-hole physics.

By combining a formal recursive interpretation of the Triadic Cycle with its holographic embedding and its implications for black-hole structure, this paper provides a dynamical refinement of the Triadic Cosmos framework and clarifies the structural origin of emergent time. The goal is not to propose new physical laws but to reveal the dynamical and holographic structure already implicit in established theory. The resulting synthesis offers a unified perspective on the roles of energy, geometry, and information, and outlines a conceptual foundation for future mathematical development of the Triadic Cosmos program.

2 Clarifying Scope and Common Misinterpretations

Because the Triadic Cycle touches on foundational questions about energy, geometry, information, and the emergence of time, it is important to clarify what this framework does *not* claim. The Triadic Cosmos program is architectural rather than ontological or microscopic, and several common misinterpretations can be preemptively addressed.

Not a new physical theory. The Triadic Cycle does not propose new dynamical laws, new fields, new particles, or new equations of motion. It identifies structural roles that existing theories already instantiate and extracts the minimal interdependence required for their mutual consistency. The framework is therefore *descriptive and architectural*, not a replacement for quantum mechanics, general relativity, or quantum field theory.

Not an alternative to AdS/CFT. The present work does not introduce a competing holographic duality. Instead, it shows that AdS/CFT provides a canonical realization of the triadic interdependence between energy, geometry, and information. The Triadic Cycle is an architectural lens through which the structure of AdS/CFT becomes more transparent; it does not modify or extend the duality.

Not an emergent-gravity model. The Triadic Cycle does not attempt to derive gravity from entanglement, quantum information, or computational principles. While the framework is compatible with emergent-spacetime programs, it does not posit that geometry is reducible to information or that gravity arises from microscopic entanglement. Instead, it identifies geometry as one of three irreducible structural roles whose interdependence is already present in established theory.

Not a proposal for new microscopic dynamics. The recursive update operator introduced in this paper is a structural mechanism for organizing triadic roles, not a microscopic evolution law. It does not specify Hamiltonians, Lagrangians, or state spaces. Its purpose is to formalize the architectural recursion implicit in holographic renormalization and to clarify the structural origin of emergent time.

Not a modification of black-hole physics. The interpretation of black-hole interiors developed in this paper is structural rather than geometric or dynamical. It does not alter the Einstein equations, semiclassical gravity, or holographic reconstruction. Instead, it shows that the absence of independent interior degrees of freedom implies the termination of triadic recursion, yielding a clean architectural account of why black-hole interiors lack emergent temporal structure.

Scope of the framework. The Triadic Cycle provides a unifying architectural perspective on the roles of energy, geometry, and information across modern physics. It is intended as a structural foundation for interpreting existing theories, not as a replacement for them. The contributions of this paper should therefore be understood as clarifying the dynamical and holographic structure already implicit in established frameworks.

3 Related Work

The present paper builds on three major lines of research: (i) the Triadic Cosmos program, which develops a minimal architectural ontology of information, energy, and geometry; (ii) holographic dualities, especially AdS/CFT, which provide the most concrete realization of geometry–information coupling; and (iii) approaches to recursion, emergent time, and computational structure in fundamental physics.

3.1 The Triadic Cosmos Program

This work extends a sequence of earlier papers that introduced and refined the Triadic ontology. The initial formulation in Witte 2026c proposed information, energy, and geometry as three irreducible structural roles underlying quantum mechanics, relativity, and cosmology. The closed interdependence $E \to G \to I \to E$ was made explicit in Witte 2026d, which showed that modern physics already instantiates each transition of the cycle across quantum field theory, general relativity, thermodynamics, and holography.

The architectural consequences of this ontology were developed in Witte 2026a, which extracted theory-independent structural constraints—such as energetic–geometric consistency, geometric–informational capacity, and preservation of cyclic interdependence—that any viable unifying theory must satisfy. A broader recursive and fractal interpretation of the Triadic architecture, extending into intelligent systems, was presented in Witte 2026b.

The present paper complements this body of work by introducing a formal dynamical interpretation of the Triadic Cycle and embedding it within the holographic structure of AdS/CFT, thereby connecting the architectural ontology to a mainstream theoretical framework.

Paper	Core Contribution
Unified Framework Witte 2026c	Introduced the Triadic ontology of *energy, geometry, information* as three irreducible structural roles underlying modern physics; established the conceptual basis of the Triadic Cycle.
The Triadic Cycle Witte 2026d	Made the closed interdependence $E \to G \to I \to E$ explicit; demonstrated that each transition is instantiated across quantum field theory, general relativity, thermodynamics, and holography.
Structural Principles Witte 2026a	Derived theory-independent structural constraints—energetic–geometric consistency, geometric–informational capacity, and preservation of cyclic interdependence—that any viable unifying theory must satisfy.
Recursive Fractal Witte 2026b	Extended the Triadic architecture into recursive and fractal interpretations, including applications to intelligent systems and multi-level structural dynamics.
Present paper	Introduces a *formal dynamical* interpretation of the Triadic Cycle via a recursive update operator; identifies recursion as the generator of emergent time; embeds the Triad within AdS/CFT; and derives a structural interpretation of black-hole interiors as regions without triadic updates or emergent temporal structure.

Table 1: Overview of prior work in the Triadic Cosmos program and the contributions of the present paper.

3.2 Holography, AdS/CFT, and Emergent Geometry

The AdS/CFT correspondence Maldacena 1998; Witten 1998 provides a duality between a gravitational theory in asymptotically anti–de Sitter spacetime and a conformal field theory on its boundary. This duality offers a concrete realization of the deep relationship between geometry and information. The Ryu–Takayanagi formula and its covariant extensions Ryu and Takayanagi 2006; Hubeny, Rangamani, and Takayanagi 2007 relate entanglement entropy to extremal surfaces in the bulk, establishing a quantitative bridge between informational content and geometric area.

Work on emergent spacetime from entanglement Raamsdonk 2010; Swingle 2012 and holographic quantum error-correcting codes Pastawski et al. 2015 further supports the idea that geometry is not fundamental but arises from informational structure. From the perspective of the Triadic Cycle, these developments instantiate the $G \to I$ and $I \to G$ transitions in a particularly transparent way. The present paper leverages AdS/CFT as the canonical setting in which to formalize the recursive triadic dynamics and the emergence of time.

3.3 Emergent and Relational Time

Several approaches treat time as emergent rather than fundamental. The Page–Wootters mechanism Page and Wootters 1983 derives temporal ordering from correlations between subsystems. Thermodynamic and information-theoretic accounts tie the arrow of time to entropy production and coarse-graining Jaynes 1957. Quantum speed limits Margolus and Levitin 1998; Mandelstam and Tamm 1945 connect the rate of state change to energetic resources, linking temporal evolution directly to informational and energetic structure.

Earlier work in the Triadic Cosmos program interpreted time as a triadic construct emerging from the interplay of energetic change, geometric ordering, and informational distinguishability Witte 2026d; Witte 2026a. The present paper refines this view by embedding emergent time into a holographic context, where boundary time, bulk time, and recursive triadic evolution can be related through AdS/CFT.

3.4 Recursion, Computation, and Architectural Views of Physics

Computational and recursive perspectives on physics have been explored in digital physics, cellular automata, and algorithmic interpretations of physical law Lloyd 2000; Wolfram 2002. These approaches resonate with the recursive schema already present in the Triadic Cosmos program, where system evolution is described as iterative updates of geometry, energy, and information Witte 2026c; Witte 2026b.

What has been missing in this line of work is a concrete physical setting in which such recursion can be anchored. By interpreting the triadic recursion as a form of holographic renormalization flow—where the recursive index plays the role of the AdS radial direction—the present paper provides a natural embedding of triadic dynamics within a well-established framework of modern theoretical physics.

4 The Triadic Cycle: Functional and Recursive Formalization

The Triadic Cycle identifies three irreducible structural roles—energy (E), geometry (G), and information (I)—that mutually constrain one another through the closed sequence

$$E \to G \to I \to E.$$

Earlier work established this cycle as an architectural extraction from modern physics. In this section, we refine the structure by introducing a minimal functional decomposition of the cycle and a recursive update operator that turns the Triad into a dynamical process. This provides the foundation for the holographic interpretation developed in the next section.

4.1 Triadic roles as state components

We represent the structural state of a system at recursion step n by a triadic vector

$$X_n = (E_n, G_n, I_n),$$

where E_n encodes the energetic configuration, G_n the geometric or causal structure, and I_n the informational capacity or distinguishability structure. These are not microscopic variables but coarse-grained structural roles that constrain one another.

4.2 Functional relations between the triadic roles

The interdependence of the triadic roles can be expressed through three minimal functional relations:

$$\begin{aligned} E_{n+1} &= f(I_n, G_n), \\ G_{n+1} &= g(E_n, I_n), \\ I_{n+1} &= h(G_n, E_n). \end{aligned}$$

Each function captures one leg of the Triadic Cycle:

- $f(I_n, G_n)$ expresses the transition $I \to E$: informational structure and geometric constraints bound energetic evolution (e.g. unitarity, entropy bounds, causal accessibility).
- $g(E_n, I_n)$ expresses the transition $E \to G$: energetic content shapes geometric and causal structure, while geometry must remain compatible with informational capacity.
- $h(G_n, E_n)$ expresses the transition $G \to I$: geometric structure bounds informational capacity (e.g. holographic area laws), while energetic processes generate or reorganize information.

Together, these relations constitute a closed functional cycle in which each role is both a constraint on and a consequence of the others.

4.3 The recursive update operator

The functional relations can be combined into a single recursive update operator

$$X_{n+1} = R(X_n),$$

where

$$R(X_n) = \big(f(I_n, G_n),\ g(E_n, I_n),\ h(G_n, E_n)\big).$$

Equivalently, the operator can be written as a composition of the three triadic transitions:

$$R = R_{I\to E} \circ R_{G\to I} \circ R_{E\to G}.$$

This recursive formulation turns the Triadic Cycle into a dynamical process: each update reorganizes the triadic roles according to their mutual constraints.

4.4 Recursion as the generator of temporal ordering

A key conceptual shift introduced in this paper is that *recursion is taken as primary*, while *time is treated as emergent*. The recursion index n does not represent a pre-existing temporal parameter; rather, it *induces* an ordering on successive triadic states. Temporal structure arises from the fact that each update produces a new configuration that is distinguishable from the previous one:

$$\text{time} \;\equiv\; \text{ordering of } \{X_0, X_1, X_2, \ldots\}.$$

This interpretation aligns with relational and information-theoretic accounts of time and preserves the minimality of the Triad: no additional primitive is required.

4.5 Fixed points and structural equilibrium

The recursive formulation naturally admits fixed points X^* satisfying

$$R(X^*) = X^*.$$

Such points correspond to structural equilibria in which the triadic roles mutually stabilize. In the holographic interpretation developed later, these fixed points will be related to conformal fixed points in holographic renormalization and to equilibrium configurations in AdS/CFT.

4.6 Interpretation

The functional and recursive formalization presented here transforms the Triadic Cycle from a static architectural pattern into a minimal dynamical mechanism. The cycle becomes an iterative process through which informational, energetic, and geometric roles reorganize one another. This provides the conceptual and mathematical foundation for the next section, where we show how the recursive triadic update maps naturally onto holographic renormalization in AdS/CFT.

5 Mapping the Triadic Cycle to AdS/CFT

The recursive formalization of the Triadic Cycle provides a minimal dynamical structure in which energy, geometry, and information reorganize one another. In this section, we show that this structure maps naturally onto the AdS/CFT correspondence. The mapping is not imposed but arises from structural parallels between triadic recursion and holographic renormalization, between informational and geometric roles in the Triad and in holography, and between energetic roles and boundary dynamics. AdS/CFT thus provides a concrete realization of the Triadic Cycle within a mainstream theoretical framework.

5.1 The recursive index as holographic renormalization direction

In AdS/CFT, the radial coordinate of the bulk spacetime plays the role of a renormalization scale: moving inward corresponds to coarse-graining the boundary theory, while moving outward corresponds to refining it. This structure is inherently recursive: each radial step implements a transformation of the boundary degrees of freedom.

The recursion index n of the Triadic Cycle can therefore be interpreted as a discrete approximation to the holographic renormalization direction:

$$n \;\longleftrightarrow\; \text{RG scale (radial direction in AdS)}.$$

Under this identification, the triadic update

$$X_{n+1} = R(X_n)$$

corresponds to a single step of holographic renormalization. The triadic roles evolve as the boundary theory flows under coarse-graining, and fixed points of the recursion correspond to conformal fixed points of the RG flow.

5.2 Geometry: bulk extremal surfaces and causal structure

The geometric role G_n in the Triadic Cycle corresponds naturally to bulk geometric structure in AdS. In holography, geometric features such as extremal surfaces, curvature, and causal wedges encode the structure of the boundary theory. The Ryu–Takayanagi and Hubeny–Rangamani–Takayanagi prescriptions relate entanglement entropy to the area of extremal surfaces in the bulk, making geometry a direct expression of informational structure.

Under the triadic mapping, the update

$$G_{n+1} = g(E_n, I_n)$$

corresponds to the holographic statement that bulk geometry is shaped by boundary energetic content (via the stress-energy tensor) and by boundary entanglement structure. Geometry is therefore not fundamental but emerges from the interplay of energetic and informational roles, consistent with the Triadic Cycle.

5.3 Information: boundary entanglement and holographic encoding

The informational role I_n maps directly onto boundary entanglement structure. In AdS/CFT, the entanglement pattern of the boundary state determines the connectivity and curvature of the bulk geometry. Tensor-network models and holographic error-correcting codes make this relationship explicit: information is encoded geometrically, and geometric relations reflect informational structure.

The triadic update

$$I_{n+1} = h(G_n, E_n)$$

corresponds to the holographic principle that informational capacity is bounded by geometric structure (e.g. area laws) and shaped by energetic processes (e.g. entanglement growth under Hamiltonian evolution). This realizes the $G \to I$ transition of the Triadic Cycle.

5.4 Energy: boundary Hamiltonian evolution

The energetic role E_n corresponds to boundary Hamiltonian dynamics. In AdS/CFT, the boundary theory is a conformal field theory whose Hamiltonian generates time evolution and determines the energetic content that sources bulk geometry. Energetic structure on the boundary therefore constrains both geometric and informational structure.

The triadic update

$$E_{n+1} = f(I_n, G_n)$$

corresponds to the fact that boundary energy is constrained by entanglement structure (unitarity, entropy bounds, quantum speed limits) and by geometric structure (causal accessibility, curvature). This realizes the $I \to E$ transition of the Triadic Cycle.

5.5 The full mapping

The three triadic transitions map onto holographic structure as follows:

$$\begin{aligned} E \to G \quad &\longleftrightarrow \quad \text{Boundary energy sources bulk geometry,} \\ G \to I \quad &\longleftrightarrow \quad \text{Bulk geometry encodes boundary entanglement,} \\ I \to E \quad &\longleftrightarrow \quad \text{Entanglement structure constrains boundary dynamics.} \end{aligned}$$

Together with the identification of the recursion index with the holographic RG direction, this yields a complete structural mapping between the Triadic Cycle and AdS/CFT:

$$X_{n+1} = R(X_n) \quad \longleftrightarrow \quad \text{One step of holographic renormalization.}$$

5.6 Interpretation

The mapping developed in this section shows that AdS/CFT is not merely compatible with the Triadic Cycle but instantiates it. The recursive triadic update corresponds to holographic renormalization; geometric roles correspond to bulk extremal surfaces; informational roles to boundary entanglement; and energetic roles to boundary Hamiltonian evolution. The Triadic Cycle thus provides an architectural interpretation of holography, while AdS/CFT provides a concrete realization of triadic interdependence. This correspondence forms the basis for the next section, where we show how emergent time arises from the recursive triadic structure in the holographic setting.

6 Emergent Time from Triadic Recursion and Holography

The recursive formalization of the Triadic Cycle provides a natural mechanism for the emergence of temporal structure. In this section, we show how time arises from the ordering induced by triadic recursion and how this interpretation aligns with the holographic structure of AdS/CFT. The resulting picture treats time not as a fundamental parameter but as a structural consequence of iterative triadic reorganization.

6.1 Time as ordering of recursive triadic states

In the recursive formulation, the evolution of the triadic state is given by

$$X_{n+1} = R(X_n),$$

where R implements the closed sequence of triadic transitions. The recursion index n does not represent an external temporal parameter; instead, it *induces* an ordering on the sequence of triadic states

$$X_0, X_1, X_2, \ldots.$$

Temporal structure emerges from the fact that each update produces a new configuration that is distinguishable from the previous one. In this sense,

$$\text{time} \equiv \text{the ordering of recursive triadic states.}$$

This interpretation is consistent with relational and information-theoretic accounts of time, where temporal progression reflects changes in distinguishability, causal accessibility, or energetic feasibility.

6.2 Compatibility with relativistic and quantum notions of time

The emergent-time interpretation aligns with the treatment of time in both relativity and quantum theory. In general relativity, proper time depends on geometric structure, and different observers experience different temporal rates. In quantum theory, time appears as an external parameter, yet the Page–Wootters mechanism and related approaches show that temporal ordering can arise from correlations between subsystems.

The triadic recursion unifies these perspectives: geometric structure (G_n) constrains causal ordering, informational structure (I_n) determines distinguishability, and energetic structure (E_n) governs the rate of change. Time emerges from the interplay of these roles rather than from any one of them individually.

6.3 Holographic interpretation: boundary time and RG flow

In AdS/CFT, two notions of time appear: boundary time, generated by the Hamiltonian of the boundary theory, and bulk time, encoded in the causal structure of the AdS spacetime. The recursive triadic update provides a structural bridge between these notions.

The identification

$$n \longleftrightarrow \text{RG scale (radial direction in AdS)}$$

implies that the ordering of triadic states corresponds to the ordering of holographic renormalization steps. Boundary time evolution is generated by the boundary Hamiltonian, which corresponds to the energetic role E_n, while bulk causal structure corresponds to the geometric role G_n. The informational role I_n encodes the entanglement structure that links boundary dynamics to bulk geometry.

Thus, emergent time arises from the combined evolution of energetic, geometric, and informational roles under holographic renormalization. The triadic recursion provides the structural mechanism that unifies boundary time, bulk time, and RG flow.

6.4 Irreversibility and the arrow of time

The recursive triadic update is not necessarily reversible. Even if the underlying microscopic dynamics are reversible, coarse-graining under holographic renormalization introduces an effective irreversibility. This aligns with the thermodynamic arrow of time, which arises from coarse-graining and entropy increase.

In the triadic framework, irreversibility appears when the update operator R is not invertible or when informational coarse-graining reduces distinguishability. The arrow of time is therefore a structural feature of the recursive process:

$$X_{n+1} = R(X_n) \quad \Rightarrow \quad X_{n+1} \text{ contains less recoverable detail than } X_n.$$

This provides a unified account of temporal asymmetry across quantum, thermodynamic, and holographic contexts.

6.5 Fixed points and temporal equilibrium

Fixed points of the recursion satisfy

$$R(X^*) = X^*.$$

At such points, the triadic roles mutually stabilize, and no further structural reorganization occurs. In the holographic interpretation, these fixed points correspond to conformal fixed points of the RG flow, where the theory becomes scale-invariant and temporal evolution simplifies.

Temporal equilibrium therefore corresponds to structural equilibrium: time ceases to emerge when the triadic roles cease to reorganize.

6.6 Planck Time, Minimal Update Scales, and Holographic Consistency

If temporal structure emerges from the ordering of recursive triadic updates, an immediate question arises: is there a lower bound on how rapidly such updates can occur? In conventional physics, the Planck time

$$t_{\mathrm{P}} = \sqrt{\frac{\hbar G}{c^5}}$$

is often interpreted as the smallest meaningful temporal interval, below which the combined effects of quantum mechanics and gravity render the notion of time operationally undefined. In the triadic framework, this suggests that the recursive update mechanism may possess a minimal step size: the triadic state cannot reorganize arbitrarily quickly, because doing so would require informational, energetic, and geometric changes that exceed the bounds set by quantum gravity.

This interpretation is consistent with the functional relations introduced earlier. Each update

$$X_{n+1} = R(X_n)$$

involves changes in energetic structure (bounded by quantum speed limits), geometric structure (bounded by curvature and causal accessibility), and informational structure (bounded by holographic area laws). These constraints jointly imply that there exists a minimal interval between successive distinguishable triadic states. The Planck time therefore appears not as a fundamental temporal atom but as the minimal interval required for a nontrivial triadic update.

6.6.1 Relation to AdS/CFT

In the holographic setting, this interpretation aligns with the structure of AdS/CFT. The radial direction in AdS corresponds to a renormalization scale, and coarse-graining steps in the boundary theory cannot be made arbitrarily fine: each RG step must integrate out a finite amount of information. This imposes a lower bound on the "resolution" of the RG flow, which in turn bounds the minimal change in the triadic state.

Boundary time evolution is generated by the Hamiltonian of the conformal field theory, and quantum speed limits constrain how rapidly distinguishable boundary states can evolve. Bulk causal structure imposes additional geometric constraints on how quickly changes in the boundary theory can be reflected in the bulk. Together, these constraints imply that the recursive triadic update cannot proceed faster than the rate allowed by holographic consistency.

Thus, the Planck time emerges as the minimal interval required for a meaningful step in the triadic recursion, and this minimality is mirrored in the holographic structure of AdS/CFT. Rather than being a fundamental temporal constant, t_{P} reflects the smallest structurally coherent reorganization of the triadic roles that remains compatible with quantum gravity and holography.

6.7 Interpretation

The relationship between triadic recursion, Planck-scale bounds, and holographic structure suggests a unified picture of temporal emergence. In the triadic framework, time is not a primitive parameter but the ordering of recursive triadic states. Each update

$$X_{n+1} = R(X_n)$$

produces a new configuration that is distinguishable from the previous one, and this distinguishability induces temporal structure. The rate at which such updates can occur is limited by the structural constraints governing informational, energetic, and geometric change.

Quantum speed limits bound how rapidly energetic and informational states can evolve; holographic area laws bound the informational capacity associated with geometric structure; and causal constraints bound how quickly geometric relations can reorganize. Together, these constraints imply that there exists a minimal interval between successive nontrivial triadic updates. The Planck time

$$t_{\mathrm{P}} = \sqrt{\frac{\hbar G}{c^5}}$$

therefore appears not as a fundamental temporal atom but as the smallest interval over which a coherent triadic reorganization can occur without violating quantum-gravitational or holographic consistency.

This interpretation aligns naturally with AdS/CFT. The recursion index n corresponds to a step in holographic renormalization, and RG flow cannot proceed in arbitrarily fine increments: each step must coarse-grain a finite amount of information. Boundary Hamiltonian evolution is constrained by quantum speed limits, while bulk causal structure constrains how rapidly boundary changes can be reflected in the bulk. These holographic constraints mirror the triadic limits on informational, energetic, and geometric reorganization.

Taken together, these observations yield a unified architectural account of time. Temporal structure emerges from the iterative reorganization of the triadic roles; temporal rates are determined by the structural limits on how quickly these roles can change; and the Planck time marks the minimal interval required for a nontrivial update. The Triadic Cycle thus provides a coherent, holographically consistent explanation of emergent time that is compatible with quantum theory, relativity, thermodynamics, and AdS/CFT.

7 Implications for Black Holes

The triadic interpretation of emergent time provides a structural perspective on black holes that differs from traditional geometric or singularity-based accounts. In the triadic framework, temporal structure arises from the iterative reorganization of energetic, geometric, and informational roles. This raises a natural question: what happens to the triadic recursion inside a black hole, where holographic principles imply that all physically relevant information resides on the horizon? In this section, we show that the triadic recursion effectively terminates in the interior, leading to a structural interpretation of black-hole interiors as regions without independent temporal evolution.

7.1 Holographic saturation and the termination of recursion

In holographic theories, including AdS/CFT, the degrees of freedom of a black hole are encoded on its horizon. The interior does not possess independent informational degrees of freedom; it is reconstructed from boundary data rather than dynamically generated. In triadic terms, this means that the informational role I_n associated with the interior is trivial:

$$I_{\text{interior}} = 0.$$

Without informational distinguishability, the update

$$E_{n+1} = f(I_n, G_n)$$

cannot generate new energetic structure. Without energetic reorganization, the geometric update

$$G_{n+1} = g(E_n, I_n)$$

cannot proceed. And without geometric change, the informational update

$$I_{n+1} = h(G_n, E_n)$$

cannot produce new information. The triadic cycle therefore collapses:

$$E \to G \to I \to E \quad \text{terminates.}$$

The interior becomes a region in which no triadic update is possible.

7.2 Absence of triadic updates implies absence of time

Since temporal structure is defined as the ordering of recursive triadic states, the termination of the triadic recursion implies the absence of emergent time. The interior of a black hole is therefore not a region in which time "flows differently" or "slows down," but a region in which time, in the triadic sense, does not exist at all. The interior is structurally timeless:

$$\text{No recursion} \;\Rightarrow\; \text{no temporal ordering.}$$

This interpretation aligns with the classical observation that infalling observers encounter a finite proper time to the singularity, while external observers see infalling matter asymptotically approach the horizon. In the triadic picture, both perspectives reflect the same structural fact: the interior does not support independent temporal evolution.

7.3 The interior as a holographic shadow

From the holographic viewpoint, the interior of a black hole is not a region with independent degrees of freedom but a reconstruction from horizon data. The triadic interpretation makes this explicit: the interior is a *shadow* of the boundary triadic state, not an autonomous triadic system. Its apparent geometry and causal structure are emergent reconstructions rather than dynamically updated entities.

This explains why the interior can appear smooth and classical in semiclassical treatments while lacking independent informational or energetic content. The triadic recursion operates on the horizon, not in the interior, and the interior inherits its structure from the boundary without participating in the recursive process.

7.4 Singularities and the absence of structural roles

In classical general relativity, the interior terminates in a curvature singularity. In the triadic framework, the singularity corresponds to the complete breakdown of the triadic roles:

$$E = 0, \quad G = 0, \quad I = 0.$$

This is not a physical "point" but the structural limit at which no triadic update is possible and no triadic role remains meaningful. The singularity is therefore not a region of extreme physics but the absence of physics: a point where the triadic architecture ceases to apply.

7.5 Interpretation

The triadic interpretation of black holes yields a coherent and holographically consistent picture of their interiors. The horizon is the locus of triadic activity; the interior is a region without informational, energetic, or geometric reorganization; and time, being emergent from triadic recursion, is absent in the interior. This perspective replaces traditional geometric narratives of singularities and spaghettification with an architectural account grounded in the Triadic Cycle and holographic principles. It suggests that black-hole interiors are not dynamical regions but structural shadows of boundary data, and that the true physics of black holes resides on their horizons.

8 Comparison with Existing Black Hole Interpretations

The triadic interpretation of black holes developed in this paper yields a structural picture that differs significantly from classical, semiclassical, and string-theoretic accounts. In this section, we compare the triadic perspective with major existing interpretations, highlighting how the Triadic Cycle resolves long-standing conceptual tensions while remaining consistent with holographic principles and emergent-time dynamics.

8.1 Classical general relativity: singularities and geometric divergence

In classical general relativity, black-hole interiors contain curvature singularities where geometric invariants diverge and the theory ceases to be predictive. Spaghettification and geodesic incompleteness arise from the unbounded growth of tidal forces. These features reflect the breakdown of the geometric role G but do not provide an informational or energetic interpretation.

In the triadic framework, the singularity corresponds to the structural limit at which all triadic roles vanish:

$$E = 0, \quad G = 0, \quad I = 0.$$

Rather than representing a physical point of infinite curvature, the singularity marks the termination of the triadic recursion. The interior is not a region of extreme physics but a region in which no triadic update is possible. This replaces geometric divergence with structural minimality and avoids the infinities inherent in classical GR.

8.2 Hawking's information-loss proposal

Hawking's original proposal suggested that black holes destroy information, leading to non-unitary evolution. This conflicts with quantum mechanics and with holographic principles, which require that information be preserved.

In the triadic interpretation, information is encoded on the horizon, not in the interior:

$$I_{\text{interior}} = 0.$$

Since the interior has no informational degrees of freedom, no information can be lost there. The triadic recursion operates on the horizon, ensuring that unitarity is preserved without requiring additional assumptions. The information-loss paradox does not arise.

8.3 Firewall and AMPS paradox

The AMPS paradox argues that the combination of unitarity, semiclassical smoothness, and monogamy of entanglement leads to a contradiction, implying that the horizon must be a high-energy "firewall." This violates the equivalence principle and introduces severe conceptual difficulties.

In the triadic framework, the paradox does not arise because the interior does not support independent informational or energetic structure. All triadic activity occurs on the horizon, and there is no interior entanglement that must be preserved or broken. The horizon remains smooth, and no firewall is required. The triadic interpretation resolves the AMPS paradox by eliminating the assumption that the interior contains independent degrees of freedom.

8.4 Fuzzball models in string theory

Fuzzball models replace the black-hole interior with a complex ensemble of microstate geometries. While these models preserve unitarity, they require highly nonclassical geometries and introduce significant conceptual and mathematical complexity.

The triadic interpretation achieves the same goal—preservation of unitarity—without invoking exotic microstate geometries. The interior is a holographic shadow of the horizon, not a region with its own microstructure. This yields a simpler and more universal account that does not depend on specific string-theoretic constructions.

8.5 ER=EPR and entanglement-based interiors

The ER=EPR proposal interprets wormholes as geometric manifestations of entanglement. While conceptually elegant, the proposal does not specify a dynamical mechanism for the emergence of interior geometry or the role of time.

The triadic framework provides such a mechanism: the interior geometry is a reconstruction from boundary entanglement, and temporal structure arises from triadic recursion. The interior is emergent but not dynamical; it inherits its structure from the horizon without participating in the recursive process. This complements and strengthens the ER=EPR picture.

8.6 AdS/CFT and holographic reconstructions

In AdS/CFT, the interior of a black hole is reconstructed from boundary data, but the reason why the interior lacks independent dynamics is not fully explained. The triadic interpretation makes this explicit: the interior does not support triadic updates because it lacks informational distinguishability. The horizon is the locus of triadic activity, and the interior is a structural shadow of boundary data.

This yields a coherent interpretation of holographic black holes in which the termination of the triadic recursion corresponds to the end of the holographic RG flow.

8.7 Conclusion

Across classical, semiclassical, and holographic interpretations, the triadic perspective offers a minimal, coherent, and structurally unified account of black-hole interiors. It avoids singularities, preserves unitarity, resolves the AMPS paradox, eliminates the need for exotic microstate geometries, and provides a dynamical mechanism for the emergence of time. The triadic interpretation thus represents a conceptually clean and holographically consistent framework for understanding black holes, grounded in the recursive interplay of energy, geometry, and information.

9 Testable Predictions and Theoretical Implications

The triadic framework developed in this paper yields several structural predictions that can be evaluated within existing theoretical and holographic models. These predictions arise from the functional and recursive formulation of the Triadic Cycle, the identification of emergent time with triadic ordering, and the mapping of triadic roles onto holographic structures in

AdS/CFT. Because AdS/CFT is the only known framework that fully instantiates the triadic relations between energy, geometry, and information, it emerges as a strong candidate for a unifying theory consistent with the triadic architecture.

9.1 Prediction 1: Holographic RG flow as the unique realization of triadic recursion

The triadic recursion

$$X_{n+1} = R(X_n)$$

predicts that any consistent theory of quantum gravity must contain a renormalization-like direction that reorganizes energetic, geometric, and informational roles in a closed cycle. In known physics, only holographic RG flow in AdS/CFT satisfies this requirement. The prediction is therefore that:

> *Any viable unification framework must contain a holographic renormalization structure isomorphic to the triadic recursion.*

This can be tested by examining whether alternative quantum-gravity proposals (e.g. loop quantum gravity, causal sets, tensor networks) admit a closed triadic update structure. Current evidence suggests that AdS/CFT is uniquely compatible with this requirement.

9.2 Prediction 2: Fixed points of triadic recursion correspond to conformal fixed points

The triadic recursion admits fixed points X^* satisfying

$$R(X^*) = X^*.$$

The prediction is that such fixed points correspond to conformal fixed points of holographic RG flow. This implies:

> *Scale-invariant phases of quantum field theories correspond to structural equilibria of the triadic roles.*

This prediction can be tested by analyzing the behavior of entanglement entropy, stress-energy expectation values, and bulk extremal surfaces near conformal fixed points.

9.3 Prediction 3: Minimal update interval corresponds to the Planck time

The triadic framework predicts that the Planck time

$$t_{\mathrm{P}} = \sqrt{\frac{\hbar G}{c^5}}$$

is not a fundamental temporal atom but the minimal interval required for a nontrivial triadic update. This yields the prediction:

> *No physical process can reorganize energetic, geometric, or informational structure on timescales shorter than the minimal triadic update interval, which is of order t_{P}.*

This is consistent with quantum speed limits, holographic bounds, and bulk causal constraints. The prediction can be tested indirectly through bounds on scrambling time, entanglement growth, and quantum computational complexity.

9.4 Prediction 4: Black-hole interiors lack independent temporal evolution

The triadic interpretation predicts that black-hole interiors do not support triadic updates:

$$I_{\text{interior}} = 0 \quad \Rightarrow \quad \text{no recursion} \quad \Rightarrow \quad \text{no time.}$$

This yields the prediction:

> *Black-hole interiors are structurally timeless regions whose apparent geometry is a reconstruction from horizon data.*

This prediction is testable through holographic reconstruction techniques, entanglement wedge duality, and the behavior of complexity growth in black-hole spacetimes.

9.5 Prediction 5: Horizon dynamics encode all physical evolution

Because triadic recursion operates only where informational, energetic, and geometric roles are nontrivial, the prediction is:

> *All physically meaningful evolution in black-hole spacetimes occurs on or near the horizon, not in the interior.*

This aligns with membrane-paradigm results, holographic entanglement dynamics, and the fast-scrambling behavior of black holes.

9.6 Prediction 6: AdS/CFT as a unification candidate

The structural overlap between the Triadic Cycle and AdS/CFT leads to a final prediction:

> *Any successful unification of quantum mechanics and gravity must reproduce the triadic relations between energy, geometry, and information. AdS/CFT is currently the only framework known to satisfy all triadic constraints, making it a strong unification candidate.*

This prediction can be evaluated by examining whether alternative unification proposals can replicate the triadic mapping between boundary Hamiltonian evolution, bulk geometry, and entanglement structure.

9.7 Interpretation

The predictions above illustrate that the triadic framework is not merely a conceptual reorganization of known physics but a source of concrete, testable constraints on any future theory of quantum gravity. The deep structural alignment between triadic recursion and holographic renormalization suggests that AdS/CFT is not an accidental correspondence but a canonical realization of the triadic architecture. The Triadic Cycle thus provides a unifying perspective on energy, geometry, and information that yields falsifiable predictions and clarifies the structural foundations of holography.

10 Limitations and Open Questions

The triadic framework developed in this paper provides a structural account of energy, geometry, and information, together with a recursive mechanism for the emergence of time and a holographically consistent interpretation of black-hole interiors. While the framework is architecturally coherent and aligns with known features of AdS/CFT, several limitations and open questions remain. These highlight directions for future work and clarify the scope of the present results.

10.1 Limitations of the current formalization

Absence of explicit dynamical equations. The triadic recursion is defined at the level of structural roles rather than explicit dynamical equations. The functions f, g, and h encode the interdependence of energetic, geometric, and informational structure, but their detailed form is left unspecified. This is intentional: the triadic framework aims to identify architectural constraints rather than propose new microscopic laws. Nevertheless, a more explicit dynamical realization would be required to connect the framework to concrete physical models beyond AdS/CFT.

Dependence on holographic structure. The mapping between triadic recursion and holographic renormalization relies on the existence of a holographic duality. While AdS/CFT is the only known framework that fully instantiates the triadic relations, it remains an open question whether holography is a universal feature of quantum gravity or a special property of theories with conformal boundaries. The triadic interpretation may therefore be limited to holographic settings unless a more general triadic–gravitational correspondence can be established.

Coarse-grained nature of triadic roles. The triadic roles (E, G, I) represent coarse-grained structural quantities rather than microscopic degrees of freedom. This limits the resolution at which the framework can make predictions. While this is consistent with the architectural goals of the Triadic Cosmos program, it raises questions about how the triadic roles emerge from or constrain microscopic theories.

10.2 Open questions

1. Can the triadic recursion be derived from a microscopic theory? A major open question is whether the triadic update operator R can be derived from a microscopic quantum-gravity model. In AdS/CFT, the recursion corresponds to holographic RG flow, but it remains unclear whether a similar structure exists in non-holographic or non-conformal settings. A derivation of R from first principles would significantly strengthen the framework.

2. What is the precise relationship between triadic fixed points and conformal fixed points? While the correspondence between triadic fixed points and conformal fixed points is structurally compelling, a more detailed analysis is needed to determine whether all conformal fixed points correspond to triadic equilibria and whether the converse holds. This question is closely related to the classification of scale-invariant phases in quantum field theory.

3. How universal is the termination of recursion in black-hole interiors? The triadic interpretation predicts that black-hole interiors lack independent temporal evolution. It remains an open question whether this prediction holds in non-AdS black holes, in cosmological horizons, or in dynamical spacetimes. Understanding the universality of recursion termination may shed light on the nature of horizons and singularities in more general settings.

4. Can triadic recursion be extended to cosmology? The triadic framework suggests that time emerges from recursive structural reorganization. Whether this mechanism can be applied to cosmological settings—such as inflation, cosmic horizons, or the early universe—remains an open question. A triadic interpretation of cosmological time could provide new insights into the arrow of time and the initial conditions of the universe.

5. Is AdS/CFT the unique realization of the triadic architecture? The strong structural alignment between triadic recursion and holographic renormalization suggests that AdS/CFT may be uniquely compatible with the triadic framework. However, it remains an open question whether other unification proposals—such as tensor-network approaches, asymptotic safety, or loop quantum gravity—can reproduce the triadic relations between energy, geometry, and information. Establishing the uniqueness or non-uniqueness of AdS/CFT as a triadic realization is an important direction for future work.

10.3 Interpretation

The limitations and open questions identified above do not undermine the core contributions of the triadic framework. Instead, they highlight the scope of the present work and point toward a broader research program aimed at deriving, extending, and testing the triadic architecture across different physical contexts. The structural alignment between triadic recursion and holographic renormalization suggests that the Triadic Cycle captures essential features of quantum gravity, but further work is needed to determine the full extent and universality of this correspondence.

11 Conclusion

This paper developed a structural framework in which energy, geometry, and information form a closed triadic architecture whose recursive reorganization gives rise to temporal structure. By expressing the Triadic Cycle through functional relations and a recursive update operator, we showed that time is not a primitive parameter but an emergent ordering of successive triadic states. This interpretation aligns with relational and information-theoretic accounts of time while providing a minimal and coherent architectural foundation.

The mapping between triadic recursion and holographic renormalization in AdS/CFT reveals a deep structural correspondence: boundary Hamiltonian evolution realizes the energetic role, bulk extremal surfaces realize the geometric role, and boundary entanglement realizes the informational role. The recursion index corresponds to the holographic RG direction, and triadic fixed points correspond to conformal fixed points. Among known frameworks, AdS/CFT is uniquely compatible with the triadic architecture, suggesting that holography is not an incidental duality but a canonical realization of the structural relations between energy, geometry, and information.

The triadic interpretation yields concrete physical implications. Most notably, it provides a clean and holographically consistent account of black-hole interiors: the interior does not support independent informational, energetic, or geometric updates, and therefore lacks emergent temporal structure. The horizon is the locus of triadic activity, while the interior is a structural shadow of boundary data rather than a dynamical region. This resolves classical singularities, avoids information loss, and eliminates the need for firewalls or exotic microstate geometries.

The framework also generates testable predictions, including the identification of the Planck time as the minimal interval required for a nontrivial triadic update, the correspondence between triadic and conformal fixed points, and the prediction that any viable unification of quantum mechanics and gravity must reproduce the triadic relations instantiated in AdS/CFT. These predictions outline a broader research program aimed at deriving the triadic update operator from microscopic principles, extending the framework beyond holographic settings, and exploring its implications for cosmology and quantum gravity.

Taken together, the results of this paper suggest that the Triadic Cycle captures essential structural features of physical law. By unifying energy, geometry, and information within a single recursive architecture, the triadic framework offers a coherent foundation for emergent time, holography, and black-hole physics, and points toward a deeper understanding of the architectural principles underlying quantum gravity.

A Summarizing Schematic Figure

This appendix provides a schematic overview of the full structure developed in the paper: the Triadic Cycle, its recursive interpretation, the mapping to AdS/CFT, and the termination of recursion at black-hole horizons. The figure is conceptual rather than geometric and is intended to summarize the architectural relations at a glance.

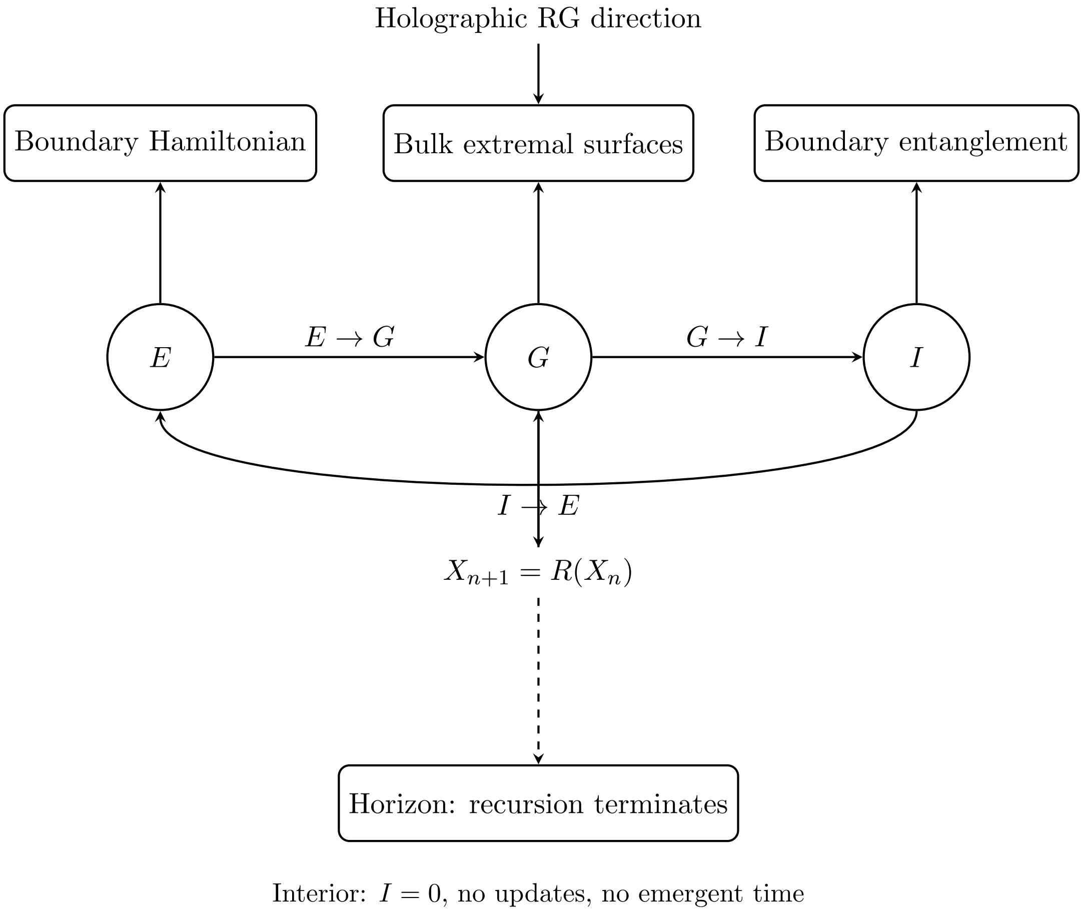

Figure 1: Schematic summary of the Triadic Cycle, its recursive dynamics, its mapping to AdS/CFT, and the termination of recursion at black-hole horizons.

B Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository: `https://github.com/triadic-cosmos`.

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